

Data indicate that sea level rise projections are too low by at least half

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1. abstract

- Roughly 1 million more people are exposed to the risks of displacement for every centimeter of global mean sea level rise (SLR). Annual costs of coastal flood damage increase by roughly ***X*** USD/cm SLR.
- Sea levels are currently rising ≈ 4.5 cm/decade and will continue to accelerate for decades under all realistic warming scenarios.
- This places forecasts, or projections, of SLR among the most consequential and value outputs in the history of science. However, the uncertainties in SLR projections are unquantified but range a minimum of more than 1.5 m (>100 million potential refugees) according to the latest assessment of the IPCC.
- SLR projections are dominated by complex models with millions of degrees of freedom and dozens of uncertain parameters and parameterizations, constrained by data whose information content is vanishingly small relative to the demands of the models.
- The outputs of the current generations of have been shown to consistently fail all comparisons with data, leaving the world without reliable information for planning and adaptation to SLR
- The broader forecasting literature has shown that increasing model complexity often degrades forecasting skill and the general rule that forecast models should be no more complex than the data can support.
- Here, we begin the construction of a data-driven, causal SLR forecast model that starts from the most naive forecast – simple data extrapolation that provides a baseline and skill over approximately 20 years – to an SLR component-level physics-informed hierarchical Bayesian calibrated against dozens of independent climatological and glaciological data sets.

2. Introduction

- Overview of SLR
- Current state of SLR projections
- Universal forecasting principles
- Failures and limitations of current SLR forecasting models

3. Data and Methods

- Extrapolation from Hamlington et al quadratic fits

- SLR semi-empirical models – Bayesian rate-state: quadratic and state relaxation time; emergence of quadratic from change in dominant SLR source
- SLR component-wise decompositions using physics-informed, hierarchical Bayesian

4. Results

- Naive forecast - data extrapolation (brief bc it's been published);
 - baseline estimation
 - likely lower-bound due to continued warming
- SLR quadratic and state relaxation time;
 - emergence of quadratic from change in dominant SLR source;
 - unlikely to be reliable by themselves (include analysis) but
 - resulting forecasts likely bracket (upper bound) truth
- SLR component-wise decompositions
 - component-wise fits; priors + posteriors
 - forecasting skill
 - Arguably most-reduced framework that faithfully represents physics
- WAIS uncertainty – Consider if simply using Bamber SEJ is appropriate here bc it's easy, honest, transparent, justifiable – can followup with a separate paper on WAIS uncertainty
- Uncertainty budget and value of information

5. Discussion

- Discuss results for SLR projections
- Forecasting skills
- Comparison with IPCC results – fall below observed trends even for warming scenarios; fail to capture basic sensitivities observed in data; component-wise comparisons all fail tests against data

6. Conclusions

1 Introduction

Overview of SLR. Global mean sea level has risen approximately 20 cm since 1900, with the rate of rise accelerating from $\sim 1.4 \text{ mm yr}^{-1}$ over the 20th century to $\sim 4.5 \text{ mm yr}^{-1}$ in the most recent decade (*Frederikse et al., 2020; Hamlington et al., 2020*). This acceleration reflects the combined effect of ocean thermal expansion and increasing mass loss from glaciers and the Greenland and Antarctic ice sheets. For each centimeter of global mean sea level rise, roughly one million additional people are exposed to the risk of coastal flooding and displacement (*Kulp and Strauss, 2019*). Sea level rise is among the most consequential and irreversible impacts of climate change: once ice is lost and oceans warmed, the commitment persists for centuries to millennia regardless of subsequent emissions reductions (*Clark et al., 2016*).

Current state of SLR projections. The Intergovernmental Panel on Climate Change (IPCC) Sixth Assessment Report (AR6) provides sea level projections based primarily on process-based ice sheet models (the Ice Sheet Model Intercomparison Project, ISMIP6; *Seroussi et al., 2020*), emulated glacier models (GlacierMIP; *Marzeion et al., 2020*), and coupled climate model output for thermal

expansion. These projections form the basis for coastal adaptation planning worldwide. The AR6 medium-confidence projections for 2100 range from approximately 0.3 m (SSP1-2.6) to 0.7 m (SSP5-8.5) relative to a 1995–2014 baseline, with a “low-likelihood, high-impact” upper bound of 2 m that reflects deep uncertainty in ice sheet dynamics (*Fox-Kemper et al.*, 2021). The gap between the medium-confidence projections and the acknowledged upper bound—a factor of three—reflects not measurement uncertainty but unquantified structural uncertainty in the process models that inform the central estimates.

Universal forecasting principles. The broader forecasting literature, developed across economics, meteorology, and public health, has established principles that are relevant to sea level prediction but have not been applied systematically. First, forecast models should be no more complex than the information content of the calibration data can support; increasing complexity beyond this threshold degrades out-of-sample skill (*Green and Armstrong*, 2015). Second, forecast combinations—weighted averages of multiple independent methods—consistently outperform individual methods, including the most sophisticated among them (*Makridakis et al.*, 2020). Third, model skill should be demonstrated through out-of-sample validation, not in-sample fit statistics. Current sea level projections rely on process models with millions of degrees of freedom and dozens of uncertain parameterizations, constrained by observational records whose information content is small relative to the demands of the models. The forecasting literature suggests that this regime is likely to produce overconfident, poorly calibrated predictions.

Failures and limitations of current SLR forecasting models. Multiple lines of evidence indicate that IPCC process-model projections systematically underestimate the rate and magnitude of sea level rise. ISMIP6 ice sheet models diverge from IMBIE satellite observations almost immediately, with most models producing ice sheets that gain mass under warming rather than losing it (*Aschwanden et al.*, 2021; *Fricker et al.*, 2025). Under continued warming to 4 °C or more, observed trends in global mean sea level exceed IPCC medium-confidence projections for all emission scenarios (*Hamlington et al.*, 2025). Known sources of low bias include: the universal use of Glen’s flow law exponent $n = 3$ when observations support $n \approx 4$, which underestimates ice sheet response by 21–35% (*Millstein et al.*, 2022; *Martin et al.*, 2026); the absence of calving, subglacial hydrology, and ice damage from most models (*Aschwanden et al.*, 2021); the sensitivity of projections to calibration methodology, with different data constraints producing factor-of-two differences in projected loss rates from the same model (*Goldberg et al.*, 2026); and incomplete uncertainty quantification that omits parametric uncertainty, which alone may exceed inter-model spread (*Aschwanden et al.*, 2019). These are not speculative concerns—they are documented, quantified biases in the models that inform the central estimates used for coastal planning.

In this work, we construct a data-driven, physics-informed hierarchical Bayesian framework for sea level prediction that does not depend on process models and that makes its assumptions, calibration range, and extrapolation uncertainty explicit. The framework proceeds in three steps of increasing physical complexity: (1) observational extrapolation of satellite-era trends, which provides a baseline lower bound; (2) an aggregate semi-empirical model relating the rate of sea level rise to global mean surface temperature, which captures the observed acceleration but conflates physically distinct components; and (3) a component-level decomposition that fits separate rate–temperature relationships to each contributor (thermal expansion, glaciers, Greenland ice sheet, Antarctic sub-components), identifies where the aggregate acceleration originates, and produces conditional predictive distributions with a decomposed uncertainty budget. Throughout, we calibrate exclusively against observational data, use Bayesian inference to propagate parametric uncertainty, and validate against independent datasets withheld from calibration.

2 Methods

2.1 Naive forecasts

We use Hamlington et al results ***Brent comment: needs fleshing out***

2.2 Semi-empirical forecasts

2.3 Physics-informed forecasts

2.3.1 Thermosteric

2.3.2 Glaciers

Mountain glaciers contribute to sea level rise through mass loss driven by surface energy balance changes under warming. We model the glacier contribution using the same rate–temperature framework as the thermosteric component, but with priors that permit moderate nonlinearity to accommodate the expected response as warming exposes new ice area and accelerates frontal ablation.

The instantaneous rate of glacier mass loss is modeled as

$$\dot{H}_{\text{gl}}(t) = a_{\text{gl}} T(t)^2 + b_{\text{gl}} T(t) + c_{\text{gl}}, \quad (1)$$

where $T(t)$ is the global mean surface temperature anomaly relative to the 1995–2005 mean, a_{gl} captures any nonlinear acceleration in the temperature sensitivity of glacier mass loss, b_{gl} is the linear mass-loss sensitivity to temperature, and c_{gl} is a background trend representing mass loss under pre-industrial equilibrium conditions (expected to be small but nonzero, as many glaciers were already adjusting to post–Little Ice Age warming).

The cumulative glacier sea level contribution is

$$H_{\text{gl}}(t) = a_{\text{gl}} I_2(t) + b_{\text{gl}} I_1(t) + c_{\text{gl}} I_0(t) + H_0^{\text{gl}}, \quad (2)$$

where $I_2(t) = \int_{t_0}^t T(\tau)^2 d\tau$, $I_1(t) = \int_{t_0}^t T(\tau) d\tau$, and $I_0(t) = t - t_0$ are the design vectors precomputed from the monthly temperature record, and H_0^{gl} is the initial sea level offset at the baseline epoch.

Priors: We assign the following priors to the glacier parameters:

$$a_{\text{gl}} \sim \mathcal{N}\left(0, 1.82 \text{ mm yr}^{-1} \text{ }^\circ\text{C}^{-2}\right), \quad (3)$$

$$b_{\text{gl}} \sim \text{HalfNormal}\left(\sigma = 5 \text{ mm yr}^{-1} \text{ }^\circ\text{C}^{-1}\right), \quad (4)$$

$$c_{\text{gl}} \sim \mathcal{N}(0.3, 2.0 \text{ mm yr}^{-1}), \quad (5)$$

$$\sigma_{\text{extra}}^{\text{gl}} \sim \text{HalfNormal}(\sigma = 2 \text{ mm}), \quad (6)$$

$$H_0^{\text{gl}} \sim \mathcal{N}(0, 10 \text{ mm}). \quad (7)$$

Unlike the thermosteric and Greenland components, the prior on a_{gl} is symmetric about zero rather than restricted to positive values. Glaciers are a finite, depleting reservoir: under sustained warming, lower-elevation glaciers disappear entirely while high-elevation accumulation zones receive increased precipitation, causing the aggregate mass-loss rate to peak and then decline. This reservoir depletion and the associated hypsometric feedback (glacier retreat to higher, colder elevations) would manifest as $a_{\text{gl}} < 0$, which a one-sided prior would exclude by construction. The scale is set so that $P(|a_{\text{gl}}| > 3 \text{ mm yr}^{-1} \text{ }^\circ\text{C}^{-2}) = 0.10$, providing weak regularization toward the linear model while allowing the data to determine the sign and magnitude of any nonlinearity. The half-normal prior on b_{gl} enforces

positive sensitivity: glaciers lose mass in a warming climate. The prior on c_{gl} is centered near zero with sufficient width to accommodate a positive background trend from post–Little Ice Age disequilibrium.

Data and calibration: The model is calibrated against the GlaMBIE global glacier consensus estimate (2000–2023; *Hock et al.*, 2024), which combines glaciological field measurements, satellite altimetry, DEM differencing, and GRACE/GRACE-FO gravimetry across all 19 glacier regions of the Randolph Glacier Inventory. We use GlaMBIE rather than the glacier component of the *Frederikse et al.* (2020) budget reconstruction because the latter relies on extrapolating a small sample of directly measured glaciers to the global total via area–volume scaling for the pre-satellite era (1961–2000), introducing model dependence that we seek to avoid. The GlaMBIE record is shorter (24 years versus 57) but is entirely observational: every annual estimate is derived from direct measurements with near-global coverage. This is a critical distinction from the Greenland SMB data (Section 2.3.3), where the observational record is itself the output of a regional climate model (RACMO); for glaciers, the rate–temperature fit extracts an empirical sensitivity directly from observations without a process-model intermediary. The observation model is

$$H_{\text{gl}}^{\text{obs}}(t_i) \sim \mathcal{N}\left(H_{\text{gl}}(t_i), \sigma_{\text{obs}}(t_i)^2 + \left(\sigma_{\text{extra}}^{\text{gl}}\right)^2\right), \quad (8)$$

where $\sigma_{\text{obs}}(t_i)$ is the time-varying observational uncertainty from the GlaMBIE consensus and $\sigma_{\text{extra}}^{\text{gl}}$ is a learned model-inadequacy term that absorbs any systematic misfit between the rate–temperature model and the observations not accounted for by measurement uncertainty (e.g., missing covariates, wrong functional form, temporally correlated residuals). This parameter measures the detectable component of structural uncertainty within the calibration window; it does not bound the extrapolation error at temperatures outside the calibration range, which is a distinct and generally larger source of uncertainty in conditional predictions. Posterior samples are drawn using an affine-invariant ensemble MCMC sampler.

Volume cap: Glaciers are a finite ice reservoir with a total volume of approximately 0.32 m SLE (*Farinotti et al.*, 2019). When projecting the fitted rate–temperature relationship forward under warming scenarios, we impose a hard ceiling: the cumulative glacier contribution to sea level cannot exceed the total available glacier volume. This is applied per Monte Carlo sample, so the conditional predictive distribution at each future time step reflects the physical constraint that mass loss must cease when no ice remains. This cap is distinct from the hypsometric feedback (glacier retreat to higher elevations reducing the effective temperature sensitivity), which has been documented at the individual glacier scale (*Marzeion et al.*, 2014) but has not been demonstrated as a globally significant effect over the observational period. The symmetric prior on a_{gl} allows the data to detect a declining sensitivity ($a_{\text{gl}} < 0$) if the hypsometric feedback is operating at the global aggregate level; the posterior is centered near zero ($a_{\text{gl}} = -0.12 [-0.60, 0.37] \text{ mm yr}^{-1} \text{ }^\circ\text{C}^{-2}$), indicating that the 2000–2024 data do not require nonlinearity in either direction.

Validation: The calibrated glacier model is validated against the glacier component of the *Frederikse et al.* (2020) budget reconstruction (1900–2018) and the satellite-era glacier component from *Horwath et al.* (2022) (1993–2016), both rebased to match GlaMBIE at the 2005 baseline to remove systematic offsets from different processing chains and peripheral glacier treatment. The *Frederikse et al.* (2020) reconstruction extends further back in time than the GlaMBIE calibration window and provides a check on whether the fitted sensitivity extrapolates to pre-2000 conditions. We also compare our posterior sensitivity b_{gl} against the emergent temperature sensitivity from the PyGEM process-based glacier model (*Rounce et al.*, 2023), which reported an approximately linear relationship between cumulative glacier mass loss and global warming level across +1.5 to +4 $^\circ\text{C}$. Our data-driven sensitivity serves as an independent observational benchmark for the process model:

agreement would increase confidence in both approaches, while disagreement would indicate where the process model departs from what the observations support.

2.3.3 Greenland

Greenland contributes to sea level rise through surface mass balance (SMB) anomalies driven by atmospheric warming and ice-dynamic discharge driven by ocean thermal forcing at marine-terminating glacier margins. We treat these separately: SMB is taken directly from observations, while discharge is modeled with a physically motivated ODE.

Surface mass balance. The *Mouginot et al.* (2019) SMB record (1972–2018) is derived from the RACMO regional climate model (*Noël et al.*, 2018), which computes surface energy balance, snow-fall, refreezing, and runoff on a high-resolution grid forced by ERA reanalysis. We use this record directly as observed data rather than fitting a rate–temperature polynomial. Fitting a simpler model to RACMO output would discard information: RACMO already resolves the spatially heterogeneous processes (albedo feedback, rain/snow partitioning, refreezing capacity) that determine SMB, and a polynomial in temperature cannot recover what the regional climate model computes. This is distinct from the glacier component, where GlaMBIE (*Hock et al.*, 2024) provides direct observations without a process-model intermediary, making a data-driven rate–temperature fit the appropriate approach.

For projections, the SMB anomaly relative to the baseline period scales with temperature (*Cuffey and Paterson*, 2010, Eq. 4.8):

$$\Delta \dot{H}_{\text{smb}} = C_T \Delta T_{\text{global}} + C_{T^2} \Delta T_{\text{global}}^2, \quad (9)$$

where C_T is the linear sensitivity incorporating both increased runoff and the partial offset from Clausius–Clapeyron scaling of accumulation (*Gregory and Huybrechts*, 2006), and C_{T^2} captures the nonlinear enhancement from ablation zone expansion under sustained warming (??). Published RCM sensitivities are reported per degree of local Greenland temperature; we convert to GMST using an Arctic amplification factor of approximately 2.0. Multiple RCMs constrain the linear sensitivity to $C_T \approx -200$ to $-250 \text{ Gt yr}^{-1} \text{ }^\circ\text{C}^{-1}$ of GMST: MAR gives -100 to $-150 \text{ Gt yr}^{-1} \text{ }^\circ\text{C}^{-1}$ of local temperature (*Fettweis et al.*, 2013; ?), RACMO gives approximately $-100 \text{ Gt yr}^{-1} \text{ }^\circ\text{C}^{-1}$ of local temperature (*Noël et al.*, 2018), with HIRHAM producing values similar to MAR. We use the inter-RCM spread as structural uncertainty: under identical SSP5-8.5 forcing, end-of-century SMB ranges from -964 Gt yr^{-1} (RACMO) to -1735 Gt yr^{-1} (MAR), a factor of ~ 1.8 (?). We draw

$$C_T \sim \mathcal{N}(-200, 80^2) \text{ Gt yr}^{-1} \text{ }^\circ\text{C}^{-1} \text{ GMST}, \quad (10)$$

$$C_{T^2} \sim \mathcal{N}(-50, 30^2) \text{ Gt yr}^{-1} \text{ }^\circ\text{C}^{-2} \text{ GMST}, \quad (11)$$

and propagate this uncertainty through the Monte Carlo ensemble. The quadratic term reflects the approximately sixfold acceleration in SMB decline above $\sim 2.7^\circ\text{C}$ GMST documented in high-resolution RCM experiments (??). As an independent check, we compute a data-driven C_T from GRACE total mass balance minus satellite-era discharge (*Otosaka et al.*, 2023; *Mankoff et al.*, 2021), which yields $C_T = 177 \pm 65 \text{ Gt yr}^{-1} \text{ }^\circ\text{C}^{-1}$ of GMST ($R^2 = 0.21$, 29 annual points). The low R^2 reflects high interannual variability from weather (NAO, atmospheric blocking) rather than a poor trend estimate; the data-driven value is within 1σ of the RCM-derived central estimate and serves as an observational lower bound.

Discharge model. The cumulative Greenland sea level contribution is

$$H_{\text{GrIS}}(t) = H_{\text{smb}}^{\text{obs}}(t) + H_{\text{dyn}}(t), \quad (12)$$

where the discharge component is governed by

$$\frac{dD_{\text{eff}}}{dt} = \frac{T_{\text{ocean}}(t) - D_{\text{eff}}(t)}{\tau}, \quad (13)$$

$$H_{\text{dyn}}(t) = \gamma_{\text{atm}} I_1^{\text{Gr}}(t) + \gamma_{\text{ocean}} \int_{t_0}^t D_{\text{eff}}(\tau) d\tau + D_0 I_0(t) + H_0^{\text{dyn}}. \quad (14)$$

Here $D_{\text{eff}}(t)$ is an effective ocean thermal state that parameterizes the lagged response of calving-front position to subsurface ocean temperature $T_{\text{ocean}}(t)$ (EN4/Argo, 200–500 m, Greenland peripheral waters; *Good et al.*, 2013; *Roemmich and Gilson*, 2009). The timescale τ represents the delay between ocean warming and grounding-line retreat. The term γ_{atm} allows for residual atmospheric influence on discharge, such as enhanced submarine melting from increased subglacial meltwater discharge (*Slater et al.*, 2020); D_0 is a secular drift.

Independence of SMB and discharge. The model treats SMB and discharge as independent, additive components. Three lines of evidence support this structure. First, *King et al.* (2020) showed that Greenland’s discharge acceleration (2000–2005) was driven by calving-front retreat, with a consistent 4–5% speedup per km of retreat that has persisted independent of interannual surface melt variability. Second, dynamic thinning propagates inland from the terminus at rates of $\sim 10 \text{ m yr}^{-1}$, an order of magnitude larger than SMB-driven thinning in the ablation zone (*Felikson et al.*, 2017); these are spatially separated processes. Third, a potential SMB–discharge feedback through driving stress ($\tau_d = \rho_{\text{ice}} g h \nabla h$)—in which SMB-driven thinning via the kinematic boundary condition $\partial h / \partial t = \dot{b} - \nabla \cdot (\mathbf{u}h)$ modifies glacier dynamics—is not supported by the data. Using the per-glacier records from *Mouginot et al.* (2019) for all 219 marine-terminating glaciers, we find no significant correlation between area-normalized cumulative SMB deficit and discharge change ($r = +0.12$, $p = 0.08$; wrong sign for the feedback hypothesis) or between cumulative mass balance and discharge change ($r = -0.01$, $p = 0.93$). The meltwater–sliding feedback is further constrained by observations at land-terminating sectors showing that increased melt can produce *deceleration* through efficient subglacial drainage development (*Tedstone et al.*, 2015). Under sustained future warming, cumulative thinning may eventually become large enough to affect discharge, but this is not detectable in the 1972–2018 record.

The posterior estimate of γ_{atm} is near zero, confirming that ocean forcing dominates decadal discharge variability.

Priors and calibration. The discharge model has 6 free parameters:

$$\gamma_{\text{atm}} \sim \text{HalfNormal}(\sigma = 2 \text{ mm yr}^{-1} \text{ } ^\circ\text{C}^{-1}), \quad (15)$$

$$\gamma_{\text{ocean}} \sim \text{HalfNormal}(\sigma = 2 \text{ mm yr}^{-1} \text{ } ^\circ\text{C}^{-1}), \quad (16)$$

$$\tau \sim \text{LogNormal}(\log 10, 0.6), \quad (17)$$

$$D_0 \sim \mathcal{N}(0, 0.5 \text{ mm yr}^{-1}), \quad (18)$$

plus a model-inadequacy term $\sigma_{\text{extra}}^{\text{dyn}}$ and baseline offset H_0^{dyn} with weakly informative priors. The prior on τ (median 10 yr, 94% CI 4–25 yr) is informed by observed outlet glacier response times

(*Straneo and Cenedese*, 2015). The model is calibrated against the discharge component of *Mouginot et al.* (2019) (1972–2018, 47 annual observations), forced by EN4/Argo subsurface ocean temperature (200–500 m, 58–80°N, 75–5°W; *Straneo et al.*, 2013). The observation model is

$$H_{\text{dyn}}^{\text{obs}}(t_j) \sim \mathcal{N}\left(H_{\text{dyn}}(t_j), \sigma_{\text{dyn}}(t_j)^2 + \left(\sigma_{\text{extra}}^{\text{dyn}}\right)^2\right). \quad (19)$$

The independent *Mankoff et al.* (2021) product (1986–2023) is withheld from calibration; the 2019–2023 period provides an out-of-sample forecast test. Total Greenland mass balance (observed SMB + modeled discharge) is validated against IMBIE (*Otosaka et al.*, 2023) and *Horwath et al.* (2022).

2.3.4 Antarctica: Peninsula and EAIS

2.4 WAIS uncertainty

3 Results

3.1 Semi-empirical forecasts

3.2 Physics-informed forecasts

3.2.1 Thermosteric

3.2.2 Glaciers

The glacier model is calibrated on 24 annual GlaMBIE observations (2000–2024) spanning approximately 1 °C of GMST warming relative to the 1995–2005 baseline. The posterior distributions for the rate–temperature parameters are:

$$a_{\text{gl}} = -0.12 \text{ } [-0.60, 0.37] \text{ mm yr}^{-1} \text{ } ^\circ\text{C}^{-2}, \quad (20)$$

$$b_{\text{gl}} = 0.77 \text{ } [0.49, 1.05] \text{ mm yr}^{-1} \text{ } ^\circ\text{C}^{-1}, \quad (21)$$

$$c_{\text{gl}} = 0.59 \text{ } [0.51, 0.67] \text{ mm yr}^{-1}, \quad (22)$$

where brackets denote the 90% credible interval. The model-inadequacy term is small ($\sigma_{\text{extra}}^{\text{gl}} = 0.11 \text{ } [0.01, 0.34] \text{ mm}$), indicating that the rate–temperature model captures the GlaMBIE observations within their stated uncertainties over the calibration period. The fit achieves $R^2 = 0.998$. We emphasize that a small σ_{extra} does not imply small structural uncertainty in conditional predictions at higher temperatures: the model is calibrated over $\sim 1^\circ\text{C}$ of warming, and extrapolation to $+3$ – 4°C carries additional risk from physical processes (reservoir depletion, hypsometric feedback) that may not be detectable within the calibration range. The volume cap and the symmetric prior on a_{gl} provide partial safeguards, but the out-of-sample structural uncertainty remains the dominant source of predictive uncertainty for the glacier component at centennial horizons.

The posterior on a_{gl} is centered near zero and spans both positive and negative values (posterior probability $P(a_{\text{gl}} > 0.5) = 1.8\%$), confirming that the data do not require nonlinearity in either direction over the observational period. The linear model ($a_{\text{gl}} = 0$) is sufficient: the instantaneous rate of glacier mass loss is well described by

$$\dot{H}_{\text{gl}} \approx 0.77 T + 0.59 \text{ mm yr}^{-1}, \quad (23)$$

where T is the GMST anomaly in $^\circ\text{C}$ relative to the 1995–2005 mean. At the 2024 GMST anomaly of approximately $+1.3^\circ\text{C}$, this implies a glacier mass-loss rate of $\sim 1.6 \text{ mm yr}^{-1}$, consistent with the GlaMBIE consensus.

The background trend $c_{\text{gl}} = 0.59 \text{ mm yr}^{-1}$ represents glacier mass loss at the baseline temperature. This is physically expected: glaciers were losing mass throughout the 20th century due to post-Little Ice Age disequilibrium and the cumulative effects of prior warming.

Comparison with process models. The data-driven linear sensitivity $b_{\text{gl}} = 0.77 \text{ mm yr}^{-1} \text{ }^{\circ}\text{C}^{-1}$ provides an observational benchmark for process-based glacier models. Integrating over 85 years (2015–2100), this sensitivity implies approximately 65 mm of cumulative glacier SLR per degree of sustained warming. *Rounce et al. (2023)*, using the PyGEM process-based glacier evolution model, reported 90–154 mm cumulative glacier loss by 2100 across +1.5 to +4 $^{\circ}\text{C}$ warming levels, implying an effective sensitivity of 35–40 mm per degree. Our data-driven value is roughly 1.5–2 $\times$ the process-model value. This discrepancy could reflect the hypsometric feedback incorporated in process models (glacier retreat to higher elevations reduces future sensitivity) or errors in the process models themselves. Process-based glacier models have been calibrated at the individual glacier scale but have not been validated against independent out-of-sample predictions of global aggregate mass loss under warming. We note that our conditional predictions are bounded by the finite glacier ice volume ($\sim 0.32 \text{ m SLE}$), which limits the practical consequence of this sensitivity difference under high-warming scenarios.

Validation against independent datasets. The GlaMBIE-calibrated model is evaluated against the *Frederikse et al. (2020)* glacier reconstruction (1900–2018) and the *Horwath et al. (2022)* satellite-era budget (1993–2016), both rebased to match GlaMBIE at the 2005 baseline. Over the satellite era (2000–2018), all three datasets agree within their stated uncertainties, confirming that the fitted rate–temperature relationship is consistent across independent observational products. Before 2000, the model extrapolation tracks the Frederikse reconstruction, though Frederikse pre-2000 glacier estimates depend on area–volume scaling extrapolations and are therefore not purely observational.

3.2.3 Greenland

3.2.4 Antarctica: Peninsula and EAIS

3.3 WAIS uncertainty

4 Discussion

4.1 Implication of results for sea-level prediction

4.2 Forecasting skill

4.3 Calibration considerations

How far outside the calibration bounds does our model still have skill?

Model inadequacy vs. structural uncertainty. Each component model includes a learned parameter σ_{extra} that quantifies the systematic misfit between the model and the calibration data beyond what measurement uncertainty explains. A small σ_{extra} indicates that the chosen functional form is adequate for the calibration period and temperature range — it does not guarantee adequacy outside that range. The distinction matters for interpretation: σ_{extra} is a lower bound on structural uncertainty within sample, not an estimate of the prediction error at temperatures the model has not seen.

For the glacier component, $\sigma_{\text{extra}} \approx 0.1$ mm over 2000–2024, during which GMST warmed by $\sim 1^\circ\text{C}$. Extrapolating the fitted linear sensitivity to $+3\text{--}4^\circ\text{C}$ introduces structural risks — reservoir depletion, hypsometric feedback, threshold behavior — that are invisible to σ_{extra} because they have not yet been expressed in the data. The same caveat applies to all components: the thermosteric model is calibrated over a temperature range that is a fraction of what SSP5-8.5 projects, and the Greenland discharge ODE has been observed for only ~ 50 years of a response time $\tau \sim 10\text{--}20$ yr. These limitations are inherent to any data-driven framework and are not unique to ours; the advantage of our approach is that the calibration range and the extrapolation lever arm are explicit, whereas process models embed analogous extrapolation assumptions in their parameterizations without reporting them as such.

4.4 Extensions and next steps

This work is intended to develop a simple hierarchical, data-driven approach to sea-level rise prediction as a complement to and diagnostic for the complex process-based models that have largely dominated discussions of sea-level rise.

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